

Fully Zero-Shot Image Dehazing

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Abstract

Image dehazing, an important image restoration problem, aims to recover clear scene content from images degraded by atmospheric haze. Existing dehazing methods rely on observing the distribution of hazy images during training: supervised approaches typically depend on synthetic datasets, leading to poor generalization in real-world scenarios; unsupervised methods are constrained by the limited diversity of observed haze conditions due to the difficulty of collecting real hazy images, and fail to generalize to unseen haze types. To address these challenges, we propose the first fully zero-shot dehazing framework that is trained without any hazy images. The framework is built upon a set of representations that remain invariant across clean and hazy images to bridge the two domains, which is both theoretically derived and experimentally validated. Consequently, we formulate dehazing as a conditional generative modeling problem and train a diffusion model solely with the invariant representations of the abundant and readily available clean images. During testing, the same representations extracted from hazy images serve as the conditional input to guide the diffusion process toward the clean image distribution. Quantitative analyses verify the effectiveness of the proposed representations, and extensive experiments across various real-world hazy datasets demonstrate our framework’s remarkable generalization ability, significantly outperforming existing methods. Our code will be available at <https://github.com/superwsc/FZSD>.

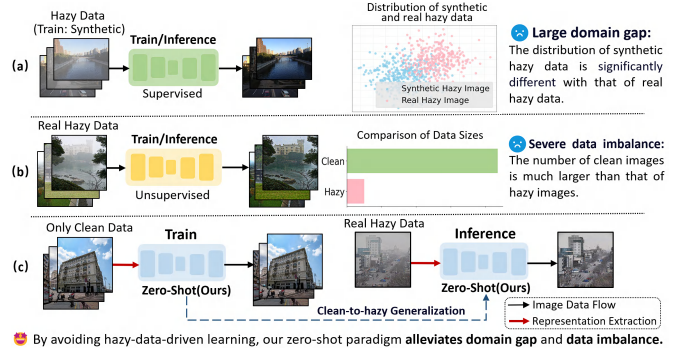


Figure 1. Comparison of our proposed zero-shot image dehazing paradigm and existing supervised and unsupervised dehazing approaches.

1. Introduction

Image dehazing is a fundamental task in low-level vision, aiming to recover clean scenes from images degraded by atmospheric scattering effects such as fog, haze, and sandstorms (He et al., 2011). It is particularly important for safety-critical applications (Li et al., 2025; Chen et al., 2025b) such as autonomous driving and video surveillance. However, dehazing poses a key challenge for learning-based image restoration methods: the real-world haze degradation is highly variable and scene-dependent, and is often unobservable during training. Existing approaches implicitly assume that dehazing models must be trained on data distributions that explicitly include hazy observations, treating hazy images as an essential driver for the learning process (Chen et al., 2024; Su et al., 2025). This assumption tightly couples model training to the availability and coverage of degraded data, thereby constraining generalization when deployed in unseen atmospheric conditions.

In practice, this haze-driven learning paradigm can be divided into two main forms, as shown in the top two rows of Fig. 1. **Supervised approaches** (Shen et al., 2025; Jin et al., 2025; Chen et al., 2024) rely on paired clean and hazy images, which are predominantly synthesized and thus fail to capture the complexity of real-world haze. **Unsupervised approaches** (Wang et al., 2025b; Su et al., 2025; Wang et al., 2025a) reduce the need for paired supervision but still require large-scale real-world hazy image datasets during

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training, whose acquisition is limited. These haze-centric unsupervised approaches underutilize the abundant priors inherent in the more readily available real clean images, and their modeling capacity is limited to the types and densities of haze observed during training. Since no dataset can exhaustively cover the full spectrum of atmospheric degradations, such methods generalize poorly when encountering unseen haze conditions during testing. Overall, existing dehazing models remain fundamentally constrained by their dependence on observing degraded data distributions. This raises an underexplored question: **is the utilization of hazy images truly necessary for learning to dehaze?**

Zero-shot paradigms (Lampert et al., 2014), as a potential solution to the aforementioned challenges, enable models trained on the source domain to generalize to the target domain that is unobserved during training. While zero-shot approaches have been widely explored in high-level vision tasks, their application to image restoration remains largely unexplored. This is mainly because restoration problems, including image dehazing, are traditionally formulated as learning mappings from degraded to clean images, implicitly assuming access to the degradation distribution during training. Moreover, establishing shared representations to bridge clean and hazy domains is challenging due to the complex and diverse nature of real-world degradations. Though a few prior works (Li et al., 2020; Kar et al., 2021) have adopted the term “zero-shot” for dehazing, these methods still rely on hazy images for training, albeit without paired clean references. Such settings should be more accurately described as unsupervised learning rather than truly zero-shot, as the target degradation distribution is still explicitly observed.

In this paper, we propose the first fully **zero-shot** dehazing framework that is trained exclusively on real clean images, as shown at the bottom of Fig. 1. We reveal that certain intrinsic features of natural scenes maintain statistical invariance across hazy and clean images, which is both derivable from physical models and confirmed by several metrics for measuring distributional similarity. By identifying and extracting these invariant features, including chromatic, structural, and luminance representations, we design a framework that maps these invariant representations to clean images without ever observing hazy data. During training, these invariant representations extracted from clean images serve as the conditional input to a generative diffusion model that is trained to reconstruct these clean images. At test time, the same model can directly process unseen hazy images, leveraging the shared representation to recover clean scenes without any previous exposure to degraded data. This approach not only circumvents the limitations of haze-dependent training but also fully exploits the diversity and richness of real-world clean images, enabling strong zero-shot generalization to previously unseen haze conditions.

Our key contributions can be summarized as follows:

- We propose the **first fully zero-shot dehazing framework** that completely decouples training from any hazy images. It is trained exclusively on real clean images and generalizes directly to unseen hazy scenes without any exposure to the degraded data.
- Several **physically grounded invariant representations** related to color, structure, and luminance are both theoretically and empirically justified, together providing a robust conditioning space for zero-shot haze removal.
- Extensive experiments show that our approach outperforms existing supervised and unsupervised methods on previously unseen real-world haze and other image degradations, highlighting the **strong generalization ability** of the proposed zero-shot paradigm.

2. Related Work

2.1. Image Dehazing

Image dehazing is a crucial task in low-level vision. Traditional approaches typically address dehazing by leveraging physical priors or statistical assumptions about hazy images. For example, He et al. (He et al., 2011) proposed a classic dehazing method based on the dark channel prior; Kim et al. (Kim et al., 2020) developed an efficient dehazing approach by estimating the saturation of the clean image.

The more recent deep-learning-based dehazing methods are primarily categorized into supervised and unsupervised approaches. Among the former, SFMN (Shen et al., 2025) proposes a spatial-frequency modulation framework that integrates cross-scale interactions and frequency-domain feature modulation. SSID (Chen et al., 2025a) employs a strategy to synthesize more realistic hazy images and generate paired datasets for training. However, this approach still fails to fully capture the distribution of all real-world hazy images. Among unsupervised methods, ZRID (Wang et al., 2025b) and DNMGDT (Su et al., 2025) incorporate traditional contrast enhancement algorithms followed by further processing through their models. This inevitably introduces artifacts into the output images. Although some works (Li et al., 2020; Kar et al., 2021; Lisani et al., 2025; Wei et al., 2023; 2022) claim to be zero-shot dehazing, they still rely on hazy images for unsupervised learning. These methods do not truly realize the capability to perform dehazing using only clean images for training.

2.2. Zero-Shot Paradigm

Zero-shot learning (ZSL) aims to acquire a model capable of performing inference on target domains that are unseen

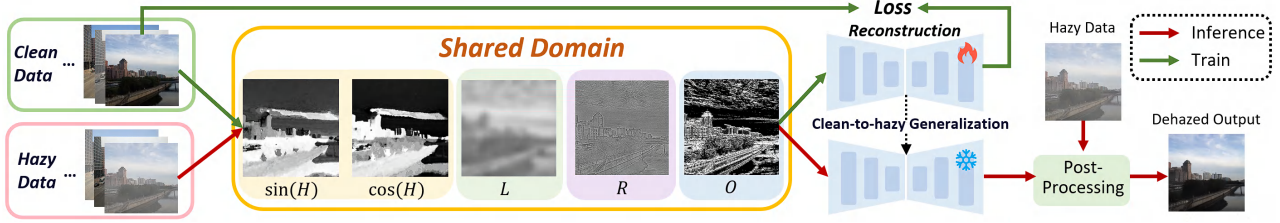


Figure 2. The overall methodology of our zero-shot dehazing framework. Our framework is trained to reconstruct clean images from domain-invariant representations and applies the same pipeline to hazy inputs by first extracting these invariant representations at test time.

during training. Early ZSL research primarily focused on recognition tasks, where semantic intermediates enable generalization beyond observed classes (Lampert et al., 2014; Palatucci et al., 2009). Building upon this foundation, ZSL has been widely explored in high-level vision tasks including image classification, object detection, and semantic segmentation, typically under unified representation learning frameworks (Xian et al., 2019; Wang et al., 2019; Pratt et al., 2023; Zhou et al., 2023; Yan et al., 2022).

Most ZSL approaches rely on constructing representations that are agnostic to specific target instances. Such abstractions are natural in high-level vision, where semantic concepts provide stable anchors across domains. In contrast, extending zero-shot principles to low-level image restoration poses challenges: restoration tasks are dominated by pixel-level degradations that are continuous and highly variable, making it difficult to identify representations that remain consistent across clean and degraded domains. A limited number of studies have attempted to extend zero-shot principles to low-level image restoration. ZeDuSR (Xu et al., 2023) proposes a zero-shot dual-lens super-resolution method that trains an image-specific SR model in test time using a single unpaired wide-angle and telephoto image. Wang et al. (Wang et al.) proposed a zero-shot low-light enhancement framework that leverages a physics-derived invariant prior. ZSVD (Varanka et al., 2025) is the first zero-shot video deraining method that removes rain without synthetic data or fine-tuning. However, no existing work has established the connection between hazy and clean images based on physical models to achieve zero-shot dehazing.

3. Methods

3.1. Motivation and Framework Overview

In the field of image dehazing, the Atmospheric Scattering Model (ASM) (Narasimhan & Nayar, 2002) serves as the fundamental bridge connecting hazy images to their corresponding clean counterparts, formulated as:

$$I^c(x) = J^c(x)t(x) + A^c[1 - t(x)], \quad (1)$$

where x is the coordinate of the pixel, and the color channel index $c \in \{R, G, B\}$. $I^c(x)$ represents the hazy image,

$J^c(x)$ corresponds to the clean image, A^c denotes the atmospheric light, and $t(x)$ represents the transmission map. For simplicity, we omit the spatial coordinate x in the remainder of this paper.

To find representations shared between hazy and clean images, our goal is to construct a mapping f such that $f(I) = f(J)$. Upon inspecting Eq. 1, we observe that the presence of the two haze-dependent variables, namely the transmission map t and the atmospheric light A^c , prevents direct invariance between I and J . To overcome this, we introduce physically reasonable assumptions and apply appropriate transformations to eliminate the influence of t and A^c from Eq. 1, thereby enabling the construction of a mapping f that yields domain-invariant representations.

The overall training and inference pipeline of our framework is illustrated in Fig. 2. We design a set of representations that remain invariant across clean and hazy images, establishing a bridge between the two domains. These representations jointly encode chromatic, structural, and luminance information, providing the model with sufficient cues for faithful image reconstruction. During training on clean images, the model simultaneously learns to reconstruct these invariant representations into high-fidelity clean images and to capture the underlying distribution of haze-free scenes. During inference, hazy images can be directly mapped to the same representations and subsequently reconstructed into clean outputs. We also employ a post-processing step (Fattal et al., 2002) to refine the saturation of the output image. Since our framework is explicitly trained to map invariant representations back to clean images, it achieves effective image dehazing without requiring any hazy training data. Considering the demand for high image quality and the fact that diffusion models excel at modeling complex data distributions, we adopt a diffusion-based framework in our approach. In the following, we provide a detailed description of each invariant representation.

3.2. Circular Chromatic Representation

To restore an image accurately, it is essential to determine the color at each position within the image to ensure the output is visually accurate and realistic. Therefore, we seek

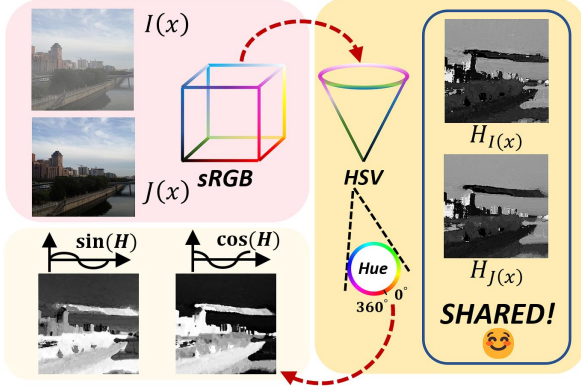


Figure 3. **Circular Chromatic Representation.** We convert the image from the RGB color space to HSV space, extract the hue component, and then apply sine and cosine functions to encode its circular structure.

a chromatic representation common to both hazy and clean images through Eq. 1. For a color image with RGB channels, we define $M = \max(R, G, B)$, $m = \min(R, G, B)$ and the color range $\Delta = M - m$. Without loss of generality, we take $M = R$ and $m = B$ as an example. We first assume a uniform airlight in each pixel (He et al., 2011): $A^R = A^G = A^B = A$. It can be observed from Eq. 1 that the relative order among the values of different color channels remains unchanged when atmospheric haze occurs. Thus, we can deduce the relationship between the color range of hazy and clean images:

$$\begin{aligned}\Delta_I &= I^R - I^B \\ &= [J^R t + A(1-t)] - [J^B t + A(1-t)] \\ &= t(J^R - J^B) \\ &= t\Delta_J.\end{aligned}\quad (2)$$

In Eq. 2, the additive term $A(1-t)$ has been successfully eliminated, leaving Δ_I and Δ_J related only by a scaling factor t . To further cancel out t , we compute the differences between two other pairs of color channels and normalize them using Δ to yield haze-invariant ratios. For example:

$$\begin{aligned}\frac{I^G - I^B}{\Delta_I} &= \frac{[J^G t + A(1-t)] - [J^B t + A(1-t)]}{t\Delta_J} \\ &= \frac{t(J^G - J^B)}{t\Delta_J} \\ &= \frac{J^G - J^B}{\Delta_J}.\end{aligned}\quad (3)$$

Thus, we have identified a chromatic representation that is invariant between hazy and clean images. To make this representation more physically explainable, we define it as follows to align with the definition of hue in the HSV color

space (Foley & Van Dam, 1982):

$$H = \begin{cases} 60^\circ \cdot \frac{G - B}{\Delta}, & \text{if } M = R, \\ 60^\circ \cdot \left(\frac{B - R}{\Delta} + 2 \right), & \text{if } M = G, \\ 60^\circ \cdot \left(\frac{R - G}{\Delta} + 4 \right), & \text{if } M = B, \\ 0^\circ, & \text{if } \Delta = 0. \end{cases} \quad (4)$$

However, hue is originally defined on a circular domain ranging from 0° to 360° , where 0° and 360° represent the identical hue. Direct linear min-max normalization, i.e., $(H - H_{min}) / (H_{max} - H_{min})$, would artificially separate adjacent hues across the $0^\circ / 360^\circ$ boundary, contradicting their inherent cyclic nature. Thus, we leverage the periodicity of sine and cosine functions to provide a natural solution to the circular nature of hue. Considering that a single sine or cosine function alone cannot establish a one-to-one mapping over the full $[0^\circ, 360^\circ]$ range, our **Circular Chromatic Representation** is comprised of two components: $\sin(H)$ and $\cos(H)$, as illustrated in Fig. 3.

3.3. Standardized Local Residual Representation

The structural information of an image captures spatial variations such as edges and textures, which are critical for faithful image reconstruction and complementary to the chromatic information introduced above. However, atmospheric haze significantly alters the appearance of an image, posing a challenge to establishing a unified structural representation across hazy and clean image domains. As shown in Fig. 4, hazy images exhibit higher brightness and lower contrast, which are respectively captured by the local mean $E(\cdot)$ and the local standard deviation $\sigma(\cdot)$. To construct a structural representation invariant to these haze-induced distortions, we first analyze $E(\cdot)$ and $\sigma(\cdot)$ of hazy and clean images.

Under the assumption that t remains approximately constant and A is uniform within a small region (He et al., 2011), we can deem $A(1-t)$ a constant in a small local patch. Moreover, the transmission map t depends solely on the distance between the scene objects and the camera sensor and is thus independent of the true radiance J . Therefore, the local mean and local standard deviation of the hazy image I can be expressed as follows:

$$E(I) = E[Jt + A(1-t)] = tE(J) + A(1-t), \quad (5)$$

$$\sigma(I) = \sigma[Jt + A(1-t)] = t\sigma(J). \quad (6)$$

To establish a structural representation that is not affected by the haze-dependent t and A , we combine Eqs. 1, 5 and 6

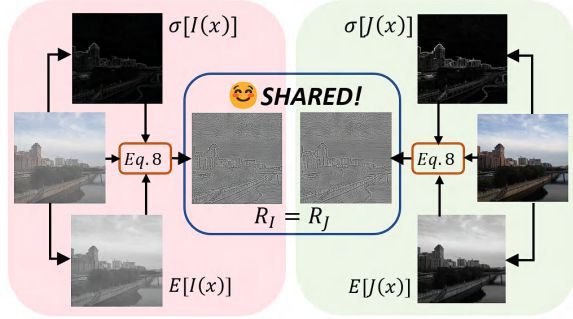


Figure 4. **Standardized Local Residual Representation.** We first compute the image’s local mean $E(\cdot)$ and local standard deviation $\sigma(\cdot)$, and then use Eq. 8 to obtain the final representation R .

and get:

$$\begin{aligned} \frac{I - E(I)}{\sigma(I)} &= \frac{[Jt + A(1 - t)] - [tE(J) + A(1 - t)]}{t\sigma(J)} \\ &= \frac{t[J - E(J)]}{t\sigma(J)} \\ &= \frac{J - E(J)}{\sigma(J)}. \end{aligned} \quad (7)$$

Eq. 7 reveals an invariant quantity related to structural information between hazy and clean images. Consequently, we define the **Standardized Local Residual Representation** as:

$$R = \frac{I - E(I)}{\sigma(I)}. \quad (8)$$

The computation process is illustrated in Fig. 4. As shown in this figure, both R_I and R_J contain rich and consistent information regarding image detail and texture, which indicates that R can serve as a structural representation to guide the image reconstruction process.

3.4. Projected Orientation Field Representation

Standardized local residual representation R captures only the local intensity variations in an image. To characterize more structural information, the gradient direction at each pixel, which encodes the orientation of intensity changes, is also required. Using this explicit directional cue, the network can more accurately reconstruct edges, textures, and other geometric structures.

To compute the image gradient direction, we first calculate the horizontal and vertical gradients of the image over a small local patch using the Sobel operator. This operator can be written as:

$$\nabla_h = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}, \nabla_v = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}. \quad (9)$$

In this subsection, we still use the previously stated assumptions, namely uniform airlight and locally constant t within a small patch. Since the coefficients of the gradient operator sum to zero, the local constant term $A(1 - t)$ is eliminated during the gradient computation. Consequently, we have:

$$\nabla_h I = \nabla_h [Jt + A(1 - t)] = t\nabla_h J, \quad (10)$$

$$\nabla_v I = \nabla_v [Jt + A(1 - t)] = t\nabla_v J. \quad (11)$$

Eqs. 10 and 11 indicate that the horizontal and vertical gradients between hazy and clean images are related by a single scaling factor t . Therefore, by taking the ratio of these gradients, we can eliminate the influence of t :

$$\frac{\nabla_h I}{\nabla_v I} = \frac{t\nabla_h J}{t\nabla_v J} = \frac{\nabla_h J}{\nabla_v J}. \quad (12)$$

This ratio effectively captures the gradient direction at each position in the image. To map the ratio in Eq. 12 to the range $(0, 1)$, we further apply absolute value and arctan transformations. Finally, our **Projected Orientation Field Representation** can be defined as:

$$O = \arctan \left[\text{abs} \left(\frac{\nabla_h I}{\nabla_v I} \right) \right]. \quad (13)$$

3.5. Low-Frequency Illumination Representation

Despite the aforementioned chromatic and structural information, the model remains unaware of luminance, leading to brightness-related artifacts. To address this, we aim to introduce another representation that explicitly encodes illumination and enables the network to recover natural results. However, significant differences exist in illumination between hazy and clean images, and the illumination information of hazy images suffers from low contrast and a lack of detail. Therefore, we cannot directly use the original image illumination as a shared representation. Luckily, image details have already been captured by the previously defined structural representations; thus, an approximate illumination distribution of the image is enough for the network.

Based on the above analyses, we use different methods to extract the **Low-Frequency Illumination Representation** of clean and hazy images. During training, we convert clean images to grayscale and apply a large-kernel Gaussian blur. To enhance the robustness of the model, we apply random gamma adjustments to the blurred result as the final representation:

$$L_{train} = \text{Gaussian}(J)^\gamma. \quad (14)$$

During inference, we handle the illumination of hazy images using two steps. First, we apply contrast-limited adaptive histogram equalization (CLAHE) (Reza, 2004) to the



Figure 5. Qualitative comparison on the RTTS and URHI (Li et al., 2019) datasets. The top two rows show results on the RTTS dataset, while the bottom two rows present results on the URHI dataset. Zoom in for better visualization.

grayscale hazy image to normalize its contrast. Subsequently, a Gaussian blur with a large kernel size is applied:

$$L_{inference} = \text{Gaussian}(\text{CLAHE}(I)). \quad (15)$$

This design ensures that hazy and clean images yield similar brightness representations, thereby guiding the reconstruction toward correct overall brightness. Further experimental validation is provided in Section 4.3 and the Appendix C.

4. Experiments

4.1. Experimental Settings

Datasets. For training, we merge the clean images from the OTS (Li et al., 2019) and Snow100K (Liu et al., 2018) datasets, yielding approximately 20,000 clean images in total. To evaluate our model both qualitatively and quantitatively, we conduct experiments on four real-world hazy image datasets: URHI (Li et al., 2019), RTTS (Li et al., 2019), 500 Foggy and Fattal dataset (Fattal, 2014).

Evaluation Metrics. As the evaluation datasets lack ground truth, all assessments are conducted using no-reference image quality metrics. We employ the FADE (Choi et al., 2015) to quantitatively measure haze density, offering an objective assessment of atmospheric scattering severity. Additionally, we use MUSIQ (Ke et al., 2021) and image entropy (EN), which reflect texture complexity and amount of information. For further perceptual evaluation, we include PAQ2PIQ (Ying et al., 2020), which predicts image quality by simulating human visual perception.

Implementation Details. Our proposed model is implemented with PyTorch 2.2.2 and trained on two NVIDIA RTX A6000 GPUs. Our network backbone is built upon Stable Diffusion v2-1 (Rombach et al., 2022). We optimize the model over 20,000 training iterations using the AdamW

optimizer (Loshchilov & Hutter, 2019) at a learning rate of 3×10^{-5} , employing a batch size of 16. We adopt the same loss functions as Stable Diffusion v2-1. The random seed is set to 231, and the resolution for both training and inference is 512×512 .

4.2. Comparisons with State-of-the-art Methods

We compare our model against 9 state-of-the-art (SOTA) methods including 6 supervised methods and 3 unsupervised methods. Fig. 5 and 6 illustrate the performance of our approach and these competing approaches. As evident from these figures, the supervised models such as DEA-Net (Chen et al., 2024) and SFMN (Shen et al., 2025) exhibit significantly degraded dehazing performance when applied to real hazy images because they tend to overfit to synthetic hazy image datasets. SSID (Chen et al., 2025a) exhibits an overall blue color cast in the first two rows of Fig. 6, because its synthetic haze method still fails to fully capture all the distribution of real-world hazy images. Unsupervised methods such as ZRID (Wang et al., 2025b) and DNMGDT (Su et al., 2025) may introduce color artifacts. For instance, the second row in Fig. 6 shows that both ZRID and DNMGDT produce significant color artifacts in the sky regions. In contrast, our model effectively mitigates these artifacts while achieving robust dehazing performance. More visual results are provided in the Appendix D.

Table 1 and 2 present the quantitative results across four datasets. Our method achieves SOTA performance in terms of the FADE metric on the URHI, RTTS, and 500 Foggy datasets, demonstrating its robust dehazing capability. Supervised models often perform poorly on FADE, which indicates their limited ability to achieve effective dehazing in real-world scenarios. Furthermore, our approach also attains SOTA results in both MUSIQ and PAQ2PIQ metrics

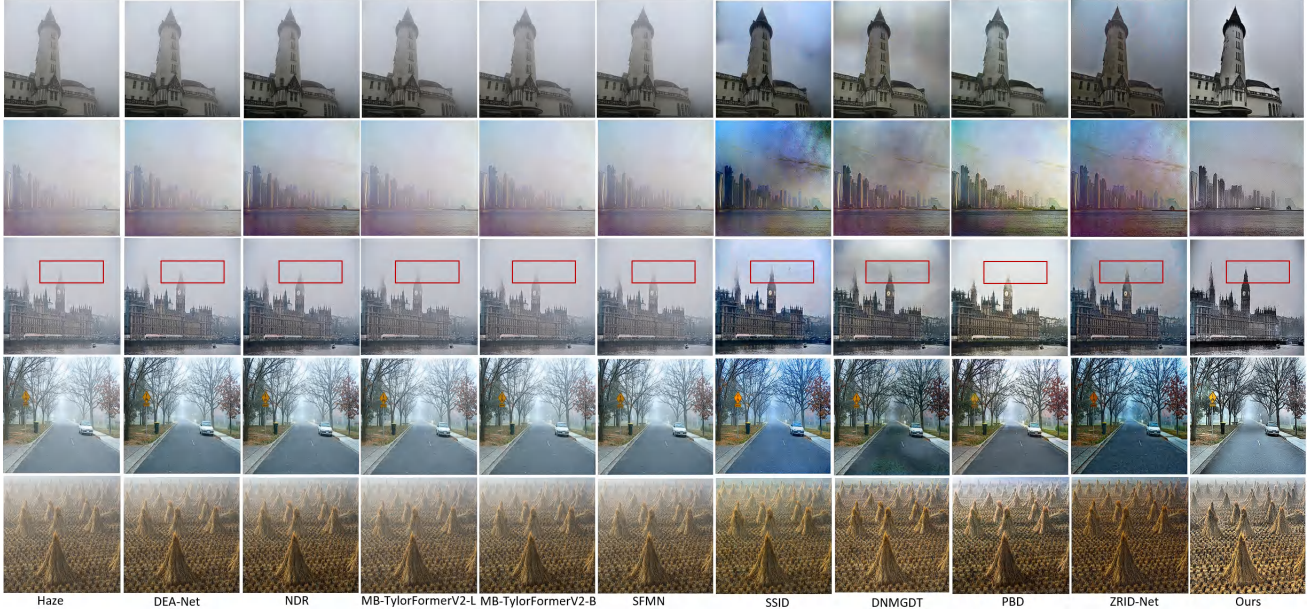


Figure 6. Qualitative comparison on the 500 Foggy and Fattal (Fattal, 2014) datasets. The top three rows show results on the 500 Foggy dataset, while the bottom two rows present results on the Fattal dataset. Zoom in for better visualization.

Table 1. Quantitative comparisons with other SOTA methods on URHI, RTTS and Fattal datasets.

Type	Models	URHI (Li et al., 2019)				RTTS (Li et al., 2019)				Fattal (Fattal, 2014)			
		FADE↓	MUSIQ↑	PAQ2PIQ↑	EN↑	FADE↓	MUSIQ↑	PAQ2PIQ↑	EN↑	FADE↓	MUSIQ↑	PAQ2PIQ↑	EN↑
Supervised	DEA-Net(Chen et al., 2024) (TIP'24)	1.751	57.09	66.75	7.132	1.740	54.58	66.36	7.087	0.669	66.06	73.19	7.327
	NDR(Yao et al., 2024) (TIP'24)	1.634	57.28	66.75	7.152	1.825	53.29	65.62	7.102	0.805	64.82	72.36	7.30
	MB-TaylorV2-B(Jin et al., 2025) (TPAMI'25)	1.729	58.47	67.11	7.103	2.027	54.23	66.09	7.032	0.714	64.59	73.13	7.314
	MB-TaylorV2-L(Jin et al., 2025) (TPAMI'25)	1.711	58.53	67.14	7.109	2.015	54.32	66.10	7.036	0.714	64.62	73.10	7.308
	SFMN(Shen et al., 2025) (TIP'25)	1.705	58.35	67.03	7.119	1.927	54.40	65.92	7.073	0.714	65.89	72.76	7.321
	SSID (Chen et al., 2025a) (TMM'25)	0.795	59.55	69.64	7.459	0.903	55.98	68.97	7.443	0.344	68.69	75.42	7.478
Unsupervised	DNMGDT (Su et al., 2025) (TMM'25)	0.754	59.89	69.68	7.347	0.897	55.80	68.80	7.358	0.417	67.72	74.74	7.382
	PBD (Wang et al., 2025a) (TCSVT'25)	0.960	58.15	67.84	7.237	1.095	53.34	66.46	7.208	0.503	63.86	72.48	7.362
	ZRID (Wang et al., 2025b) (TCSVT'25)	0.615	56.77	69.70	7.155	0.753	53.11	68.76	7.098	0.294	66.84	75.11	7.267
	Ours	0.613	62.92	71.64	7.318	0.695	59.72	71.46	7.275	0.327	70.63	75.86	7.511

The best and the second best results are highlighted in red and blue respectively.

Table 2. Quantitative comparisons with other SOTA methods on 500 Foggy dataset.

Type	Models	500 Foggy			
		FADE↓	MUSIQ↑	PAQ2PIQ↑	EN↑
Supervised	DEA-Net(Chen et al., 2024)	1.532	63.64	67.97	7.243
	NDR(Yao et al., 2024)	1.376	63.73	67.92	7.263
	MB-TaylorV2-B(Jin et al., 2025)	1.351	64.50	68.46	7.226
	MB-TaylorV2-L(Jin et al., 2025)	1.344	64.56	68.47	7.229
	SFMN(Shen et al., 2025)	1.353	64.13	68.26	7.227
	SSID (Chen et al., 2025a)	0.559	64.28	70.67	7.464
Unsupervised	DNMGDT (Su et al., 2025)	0.607	65.02	70.88	7.346
	PBD (Wang et al., 2025a)	0.882	63.72	69.42	7.322
	ZRID (Wang et al., 2025b)	0.455	62.62	70.84	7.211
	Ours	0.404	65.72	71.98	7.498

The best and the second best results are highlighted in red and blue respectively.

Table 3. The distributional similarity of the representation across hazy and clean domains.

	$\sin(H)$	$\cos(H)$	R	O	L
JS divergence ↓	0.0088	0.0136	0.0050	0.0152	0.0197
cosine similarity ↑	0.9789	0.9733	0.9897	0.9637	0.9305
Euclidean distance ↓	0.0186	0.0215	0.0097	0.0574	0.0643

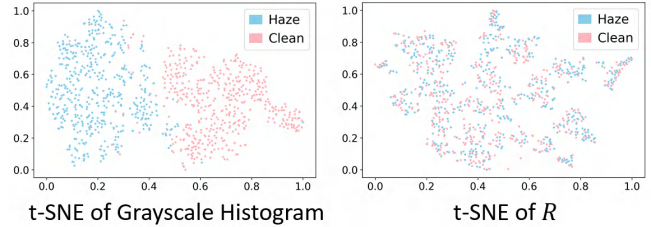


Figure 7. The t-SNE visualization of both our proposed representation R and the grayscale images of 500 hazy and 500 clean images. For each sample, we compute the histogram of its representation R and the histogram of its grayscale image, and use these histograms as input to t-SNE.

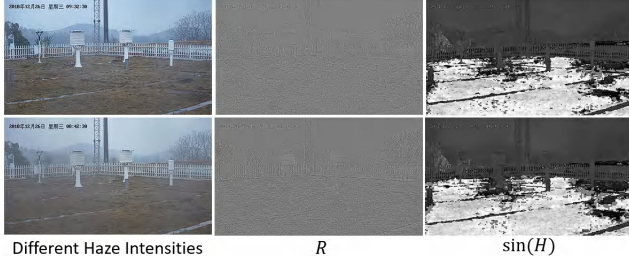
across all datasets, which indicates that our model produces images with highly informative contents and high perceptual quality as perceived by humans.

4.3. Validity of Invariant Representations

We conduct experiments from multiple aspects to validate the rationality of our designed representations. We first use



Figure 8. Ablation study of the effects of different representations.


 Figure 9. The visualization of the $\sin(H)$ and R of images depicting the same scene with varying levels of haze. Our designed representations remain largely consistent despite changes in haze intensity. Zoom in for better visualization.

500 hazy and 500 clean images to compute the histogram of each representation, then average the histograms to obtain the corresponding distributions. To quantify the distributional similarity of the representation across hazy and clean domains, we evaluate them using three metrics: Jensen-Shannon (JS) divergence, cosine similarity, and Euclidean distance. In general, two distributions can be considered highly similar when their JS divergence is below 0.05, cosine similarity exceeds 0.9, and Euclidean distance is under 0.1. As shown in Table 3, all of our proposed representations satisfy these criteria: the JS divergence is consistently below 0.02, cosine similarity exceeds 0.93, and Euclidean distance remains under 0.07 between hazy and clean images. This demonstrates that our representations are highly consistent across hazy and clean domains, providing strong quantitative validation for the effectiveness and rationality of our zero-shot dehazing paradigm.

Moreover, we illustrate the t-SNE visualization of both our proposed representation R and the grayscale images of hazy and clean domains in Fig. 7 (t-SNE visualizations of other representations are in the Appendix C). As illustrated in Fig. 7, the scatter points of the grayscale images are widely dispersed and clearly separable, whereas those of representation R are thoroughly intermixed. This demonstrates that our designed representation preserves consistent statistical characteristics across hazy and clean domains.

We further visualize the designed representations to qualitatively show their similarity. The RW-HAZE dataset (Chen et al., 2022) includes images of the same scenes with varying haze levels. We visualize the representations of these

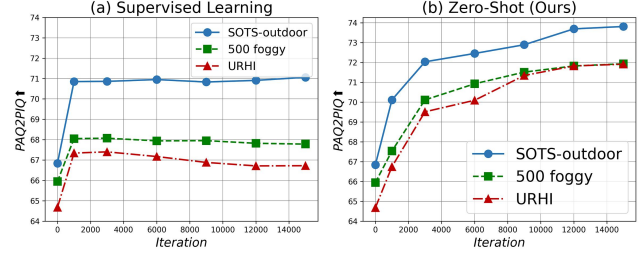


Figure 10. Comparison of training dynamics between supervised and zero-shot dehazing paradigms, illustrating the strong generalization ability of our model.

images in Fig. 9. It can be observed that these representations remain largely consistent despite changes in haze levels (more results are provided in the Appendix C). This shows that our designed representations are robust to varying degrees of haze, demonstrating the effectiveness and rationality of our representation design.

4.4. Ablation Studies

The ablation study results are shown in Fig. 8. We analyze the contribution of each representation and find that removing any one of them degrades the dehazing performance. Specifically, without $\sin(H)$ and $\cos(H)$, the model loses its ability to perceive object colors accurately, leading to incorrect color reproduction. Replacing the circular encoding with direct linear normalization of hue introduces color shifts in red regions (near $0^\circ/360^\circ$ on the hue circle) due to the violation of hue’s cyclic nature. Omitting the low-frequency illumination representation L results in outputs with mismatched brightness distributions and visible false-color artifacts. Removing the projected gradient orientation representation O impairs the model’s awareness of local intensity directions, causing residual haze. Finally, excluding the standardized local residual representation R deprives the model of essential structural cues, making it unable to faithfully reconstruct scene content. Overall, the full model incorporating all representations achieves the best performance.

4.5. Discussion

To examine the generalization behavior of different dehazing paradigms, we track the image quality during training by

periodically evaluating PAQ2PIQ on three datasets. Specifically, a supervised model DEA-Net (Chen et al., 2024) is trained on synthetic OTS dataset and evaluated on SOTS-outdoor (Li et al., 2019) as well as two unseen real-world datasets, while our zero-shot model is trained solely on clean images and evaluated on the same three datasets. As shown in Fig. 10, though the supervised model exhibits improvement on SOTS-outdoor, its performance on unseen datasets saturates or even slightly degrades as training proceeds, indicating overfitting to dataset-specific haze distributions. In contrast, our zero-shot model shows consistent and steady improvement across all datasets during training, despite never observing hazy images. This suggests that learning from clean-scene priors rather than from degradation-specific data leads the model to generalize more robustly across diverse and previously unseen haze conditions.

Besides, we further conduct a non-uniform illumination enhancement experiment to evaluate the generalization capability of our model (details in Appendix A), and a downstream object detection experiment and a downstream depth estimation experiment to validate the effectiveness of our dehazing results (details in Appendix B).

5. Conclusion

In this work, we propose the first fully zero-shot image dehazing framework trained exclusively on real clean images. By designing a set of representations from physical models that exhibit statistical invariance across hazy and clean domains, we train a generative diffusion model to learn the mapping from these invariant representations to haze-free outputs. At test time, our framework directly processes unseen hazy images by exploiting the shared representation to reconstruct clean scenes. Extensive experiments validate the effectiveness of our dehazing results, the rationality of the proposed representations, and the strong generalization capability of our zero-shot paradigm. These results demonstrate a novel and practical training strategy for image dehazing.

Acknowledgement

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Impact Statement

Due to atmospheric conditions such as suspended aerosols and moisture, image acquisition systems inevitably suffer from haze interference. Images captured under hazy con-

ditions experience a significant degradation in visibility, object contrast, and color fidelity. Hazy images can severely impair the performance of outdoor computer vision systems, including autonomous driving, surveillance, and remote sensing. Therefore, image dehazing holds substantial research significance and practical value. Our proposed framework achieves fully zero-shot image dehazing by training exclusively on clean images and delivers excellent dehazing performance. Moreover, the model exhibits strong generalization capability, enabling it to effectively tackle complex dehazing tasks. Despite its benefits, image dehazing may also entail societal risks. Excessive dependence on dehazing systems can lead to artifacts or unrealistic enhancements that distort the original scene appearance, thereby undermining perceptual reliability in applications like autonomous driving and video surveillance. To mitigate such risks, human oversight and domain expertise should be incorporated into the decision-making process.

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A. Task Generalization Results

To further validate the generalizability of our zero-shot paradigm on other degradations, we evaluate our model on an additional non-uniform illumination image enhancement task using the NPE dataset (Wang et al., 2013), without any retraining. We compare against three state-of-the-art dehazing methods: SFMN (Shen et al., 2025), SSID (Chen et al., 2025a), ZRID (Wang et al., 2025b), and one dedicated non-uniform illumination enhancement method: Zero-DCE++ (Li et al., 2022). As shown in Fig. 11 and Table 4, the compared dehazing methods produce negligible improvements under non-uniform illumination conditions, whereas our zero-shot model effectively enhances image brightness and contrast, achieving comparable quality to that of Zero-DCE++.

Table 4. Quantitative comparisons with other methods on the non-uniform illumination image enhancement task on NPE dataset.

Type	Models	NPE dataset (Wang et al., 2013)		
		MUSIQ $\uparrow$	PAQ2PIQ $\uparrow$	ENTROPY $\uparrow$
Supervised Dehazing	SFMN(Shen et al., 2025)	61.325	69.792	6.944
	SSID (Chen et al., 2025a)	61.998	71.326	7.316
Unsupervised Dehazing	ZRID (Wang et al., 2025b)	61.870	72.118	7.007
Unsupervised Non-uniform Illumination Enhancement	Zero-DCE++ (Li et al., 2022)	63.342	73.401	7.370
Zero-Shot	Ours	66.463	73.143	7.316

The best and the second best results are highlighted in red and blue respectively.

B. Downstream Object Detection and Depth Estimation Results

To assess the practical value of our dehazed results, we perform a downstream object detection experiment on the RTTS dataset using MobileNetV3-SSDLite. For example, in the first row of Fig. 12, other dehazing algorithms fail to effectively remove the haze or introduce significant artifacts, resulting in the detector being unable to identify any objects in the image. These results reveals their inability to adequately retain semantic information critical for downstream vision tasks. By comparison, feeding our restored images into the detector leads to notably better performance than using the original haze-degraded inputs.

We also perform a downstream depth estimation experiment on the RTTS dataset using Depth Anything Model (Yang et al., 2024). Compared with other methods, our dehazing results yield more accurate details in depth estimation, as shown in Fig. 13. This demonstrates that our framework can serve as an effective preprocessing step in object detection and depth estimation pipelines.

C. Validity of our Representation Design

The t-SNE visualizations of all our proposed representations and the grayscale images of 500 hazy and 500 clean images are shown in Fig. 14. For each sample, we compute the histogram of its representations and the histogram of its grayscale image, and use these histograms as input to t-SNE. As illustrated in Fig. 14, the representations' feature points from hazy and clean images are well mixed, indicating that all our representations are highly consistent across hazy and clean domains. In contrast, the scatter points of the grayscale image histogram from hazy and clean domains are widely dispersed and clearly separable. This demonstrates the effectiveness and rationality of our representation design.

To further illustrate the visual consistency of our design, we provide visualizations of the proposed representations. The RW-HAZE dataset (Chen et al., 2022) includes images of the same scenes with varying levels of haze. We visualize the representations of two scenes, as shown in Fig. 15 and 16. It can be observed that our designed representations remain largely consistent despite changes in haze levels. We also present the JS divergence for Fig. 15 and 16 in Table 5 and 6. The JS divergence of the representations is significantly reduced compared to that of images themselves, satisfying the criterion for distributional similarity. This shows that our designed representations are robust to varying degrees of haze, thereby demonstrating the effectiveness and rationality of our representation design.

D. More Visual Comparisons with Other State-of-the-art Methods

Figures 19, 20, 17, 18, and 21 show additional visual comparisons on the URHI, RTTS, 500 Foggy, Fattal, and RW-HAZE datasets, respectively. It can be observed that supervised methods often fail to fully remove haze, while unsupervised

approaches tend to introduce color artifacts. In contrast, our model achieves the best dehazing performance across all datasets.

E. Visual Comparisons on Night Scenarios and Dense Haze Scenarios

We include additional nighttime comparisons in Fig. 22 to further investigate the capability of our current method in nighttime dehazing. While our performance in certain nighttime scenarios is limited, our method still achieves comparable or even superior visual quality compared to existing state-of-the-art (SOTA) dehazing models.

We also supplement our results with real-world dense haze scenarios from the RTTS and URHI datasets in Fig. 23. Extreme dense haze poses challenges to accurate representation extraction, potentially compromising the final performance. As shown in Fig. 23, our method still outperforms or matches SOTA methods in terms of visual clarity in some dense haze scenarios. Removing dense haze is a difficult problem and we are committed to exploring more robust algorithms for dense haze in our future research.

F. Trade-off between the Dehazing Capabilities and Hallucination Suppression

Our proposed invariant representation R encodes rich structural priors, enabling the model to accurately reconstruct scene geometry and suppress generative hallucinations. As shown in the ablation study in Fig. 24, removing R may lead to hallucinations, whereas the inclusion of R ensures high-fidelity restoration.

Furthermore, to specifically evaluate geometric consistency, we analyze the structure term of SSIM between the hazy input and restored images on 500 foggy dataset, defined as $s(x, y) = \frac{\sigma_{xy} + C}{\sigma_x \sigma_y + C}$. Here, σ_{xy} represents the covariance between the hazy and restored patches, while σ_x and σ_y denote their standard deviations. As shown in Table 7, our method achieves a high $s(x, y)$ of 0.932. This score indicates that while our model restores visibility, it simultaneously maintains structural fidelity to the original scene layout. Notably, despite achieving an $s(x, y)$ score of 0.996, SFMN exhibits negligible dehazing performance. These quantitative analysis of structural fidelity confirms that R provides a rigorous geometric constraint, preventing the diffusion backbone from generating inconsistent content. These evidences validate that our framework maintains adherence to the original scene layout without hallucinating artificial details.

G. More Discussions about our Motivation

Beyond the mathematical proofs in the paper, here we further clarify the motivation and rationality behind our design. Specifically, we demonstrate why these representations are essential for high-fidelity reconstruction and how they maintain consistency across clean and hazy domains.

Circular Chromatic Representation (H). *Motivation:* To achieve high-fidelity image restoration, it is important to accurately determine the color at each position. Thus, providing a stable chromatic representation ensures the restoration process is grounded in the original scene’s chromatic characteristics. *Rationality:* According to the optical properties of scattering, haze primarily affects saturation and brightness. However, haze does not change the intrinsic color of the objects. Based on this, we posit the existence of a representation that reflects the intrinsic color of objects and remains invariant to haze. The HSV space can decouple hue from saturation and brightness, allowing hue to serve as a shared representation across the clean and hazy domains.

Standardized Local Residual Representation (R). *Motivation:* Structural information encompasses edges and fine textures and is fundamental for image reconstruction. While chromatic representations provide color consistency, they often lack fine details. Therefore, structural cues are required to ensure the restored image is faithful to the structures of the original scene. *Rationality:* Haze acts as a degradation that increases luminance and suppresses contrast, reducing the signal-to-noise ratio. Nevertheless, the fundamental scene structures (the relative spatial distribution of intensities) are only veiled but not completely obscured by haze. Therefore, we can derive an invariant structural representation by normalizing pixel values using luminance and contrast. This normalization converts signals of different strengths in clean and hazy images to a standard level, thereby producing an invariant representation across two domains.

Projected Orientation Field Representation (O). *Motivation:* While R captures structural information, its normalization leads to loss of some gradient-related information. To ensure more accurate structure integrity, more gradient information, e.g., gradient direction, is required. *Rationality:* Within small local patches, the haze can be assumed to be spatially uniform

and isotropy. This uniformity implies that while haze attenuates gradient magnitude, gradients in all directions decay by the same proportion. Consequently, we can obtain an invariant representation by calculating the ratio of the gradients in horizontal and vertical directions.

Low-Frequency Illumination Representation (L). *Motivation:* The aforementioned representations are obtained via normalizations, so the luminance information is eliminated. Without luminance constraints, the generation process may lead to unnatural illumination shifts, and relative brightness between different parts of the image may be wrong. Thus, an additional luminance feature is essential for accurate reconstruction. *Rationality:* From a physical perspective, haze increases global brightness. To bridge this domain gap, for hazy images, we employ histogram equalization-based methods to recalibrate brightness distribution. However, this may introduce artifacts. So we then apply a strong Gaussian filter to suppress these artifacts and retain only macroscopic illumination trends. For clean images, we only apply Gaussian filter. Therefore, L of both hazy and clean images are low-frequency components at a normal brightness level, and can be used as an invariant representation.

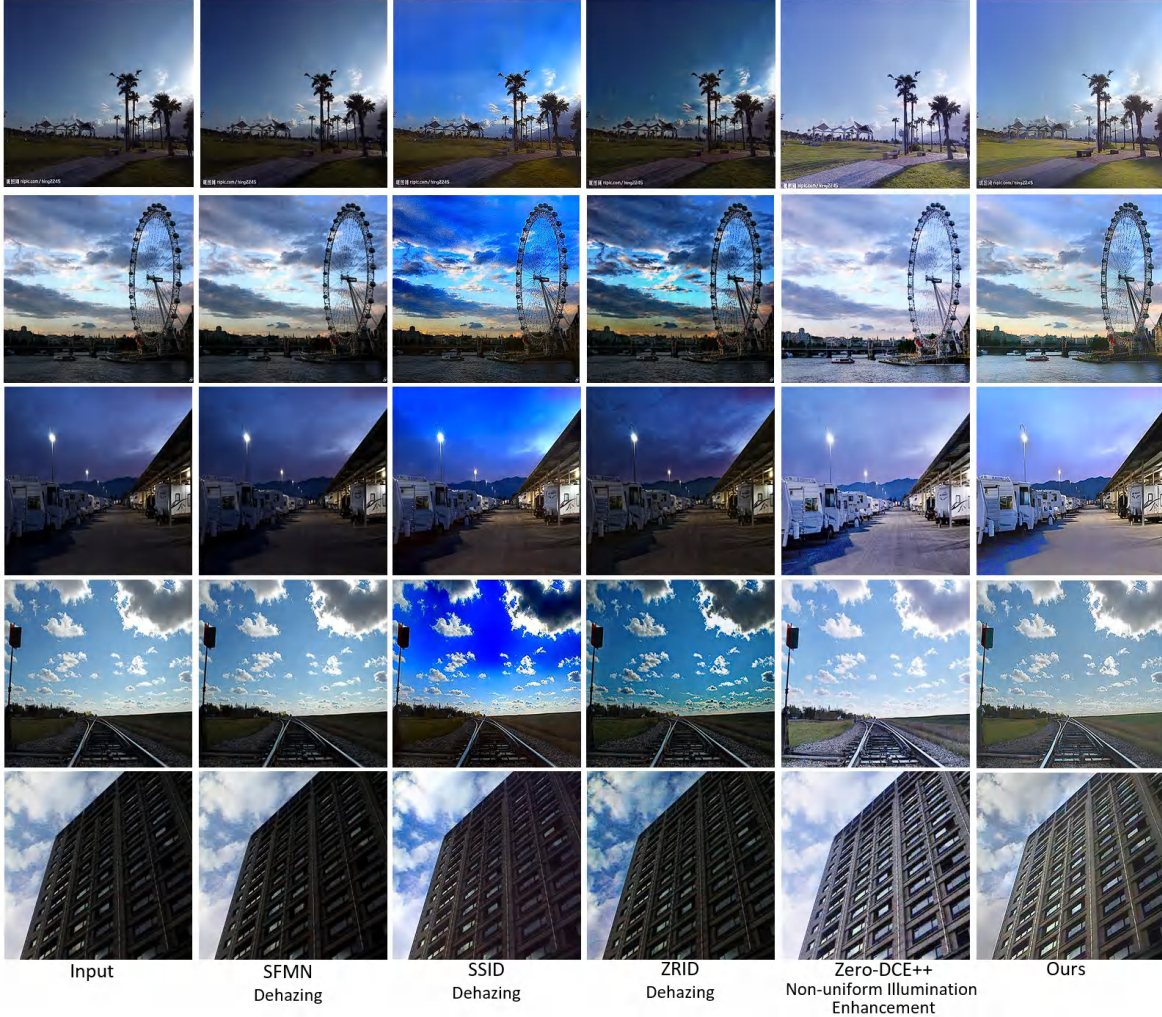


Figure 11. Visual comparisons on the non-uniform illumination image enhancement task to further validate the generalizability of our zero-shot paradigm.



Figure 12. Downstream object detection experiment on the RTTS dataset using MobileNetV3-SSDLite. Feeding our restored images into the detector leads to notably better performance.

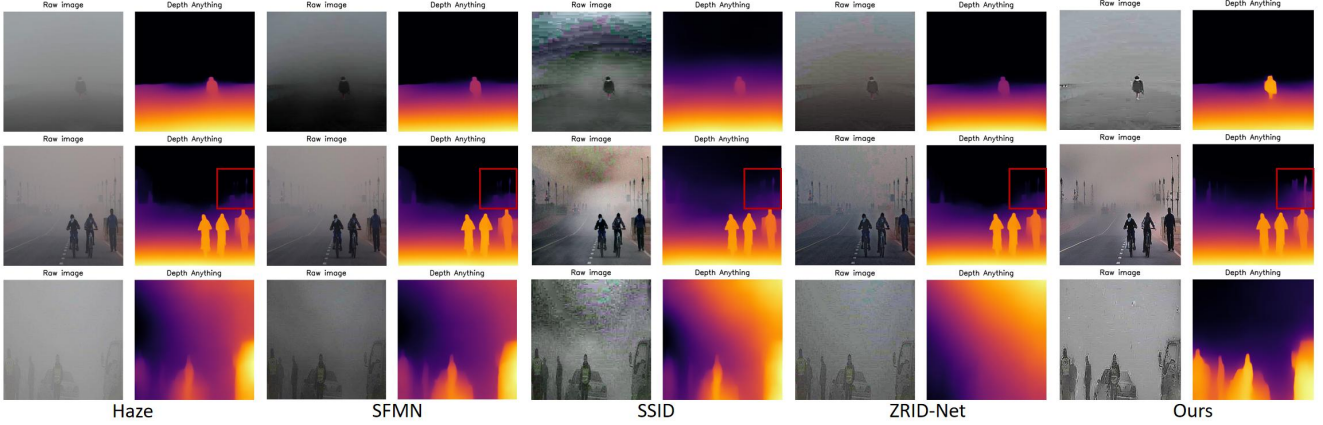


Figure 13. Downstream depth estimation experiment on the RTTS dataset using Depth Anything model (Yang et al., 2024). Feeding our restored images lead to notably better performance than others.

Table 5. The distributional similarity of the images and the representations in Fig 15.

	Image	$\sin(H)$	$\cos(H)$	R	O	L
JS divergence of the First and Sencod Row↓	0.094	0.014	0.015	0.011	0.016	0.020
JS divergence of the First and Third Row↓	0.087	0.013	0.015	0.010	0.017	0.021
JS divergence of the Second and Third Row↓	0.079	0.012	0.014	0.009	0.015	0.019

Table 6. The distributional similarity of the images and the representations in Fig 16.

	Image	$\sin(H)$	$\cos(H)$	R	O	L
JS divergence of the First and Sencod Row↓	0.083	0.010	0.012	0.007	0.014	0.020
JS divergence of the First and Third Row↓	0.078	0.013	0.013	0.009	0.015	0.019
JS divergence of the Sencod and Third Row↓	0.074	0.011	0.010	0.008	0.012	0.017

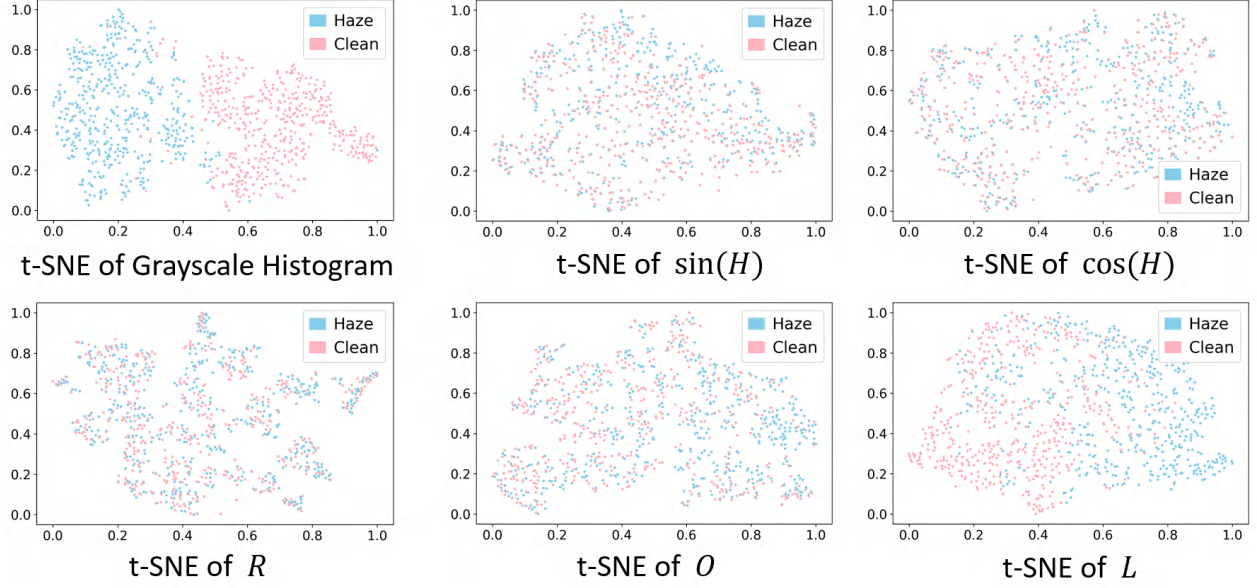


Figure 14. The t-SNE visualization of all the representations and the grayscale images of 500 hazy and 500 clean images. The feature points of all our representations from hazy and clean images are well mixed, indicating that our representations are highly consistent across hazy and clean domains. This demonstrates the effectiveness and rationality of our representation design.

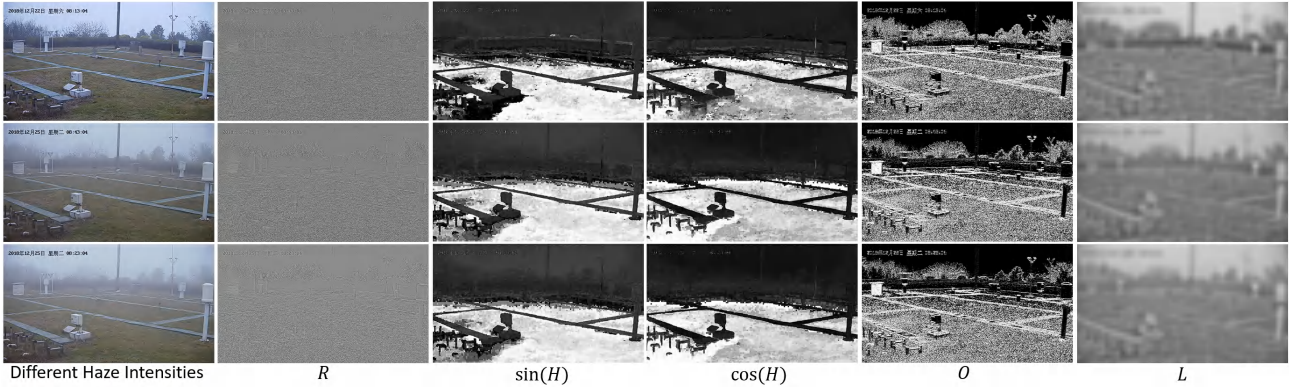


Figure 15. In the RW-HAZE dataset (Chen et al., 2022), we visualize the representations of images depicting the same scene with varying levels of haze. As shown in these visualizations, our designed representations remain largely consistent despite changes in haze intensity. Zoom in for better visualization.

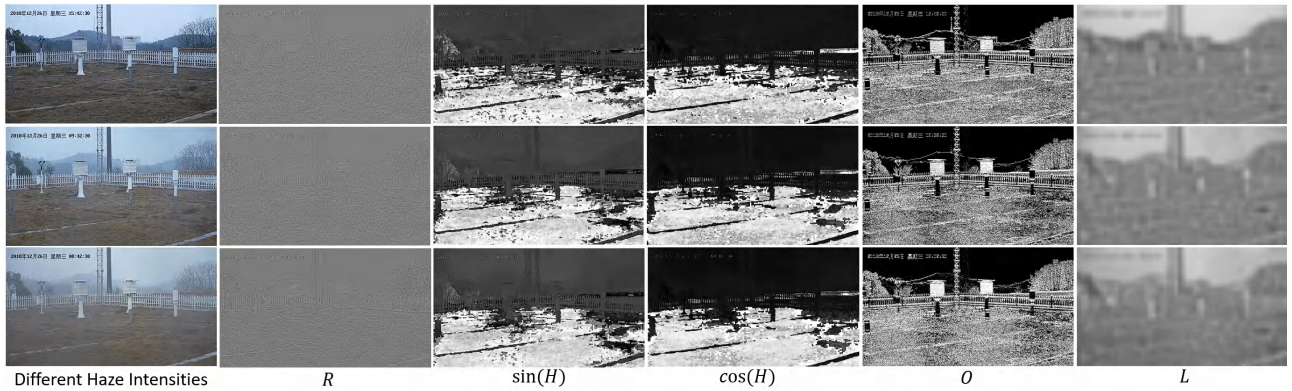


Figure 16. More visual results of our designed representations with varying levels of haze in the RW-HAZE dataset (Chen et al., 2022). Zoom in for better visualization.



Figure 17. More qualitative comparison on the 500 Foggy dataset. Zoom in for better visualization.



Figure 18. More qualitative comparison on the Fattal (Fattal, 2014) dataset. Zoom in for better visualization.

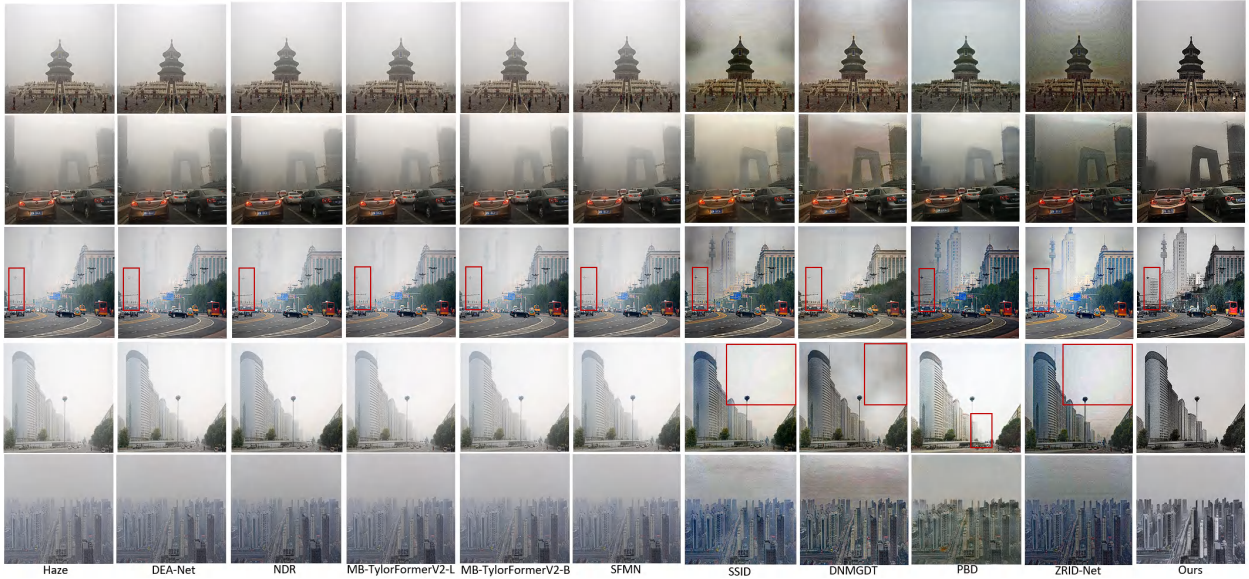


Figure 19. More qualitative comparison on the URHI (Li et al., 2019) dataset. Zoom in for better visualization.

Table 7. To specifically evaluate geometric consistency, we analyze the structure term $s(x, y)$ of SSIM between the hazy input and restored images on 500 foggy dataset.

	Ours	SSID	PBD	SFMN
$s(x, y) \uparrow$	0.932	0.930	0.920	0.996

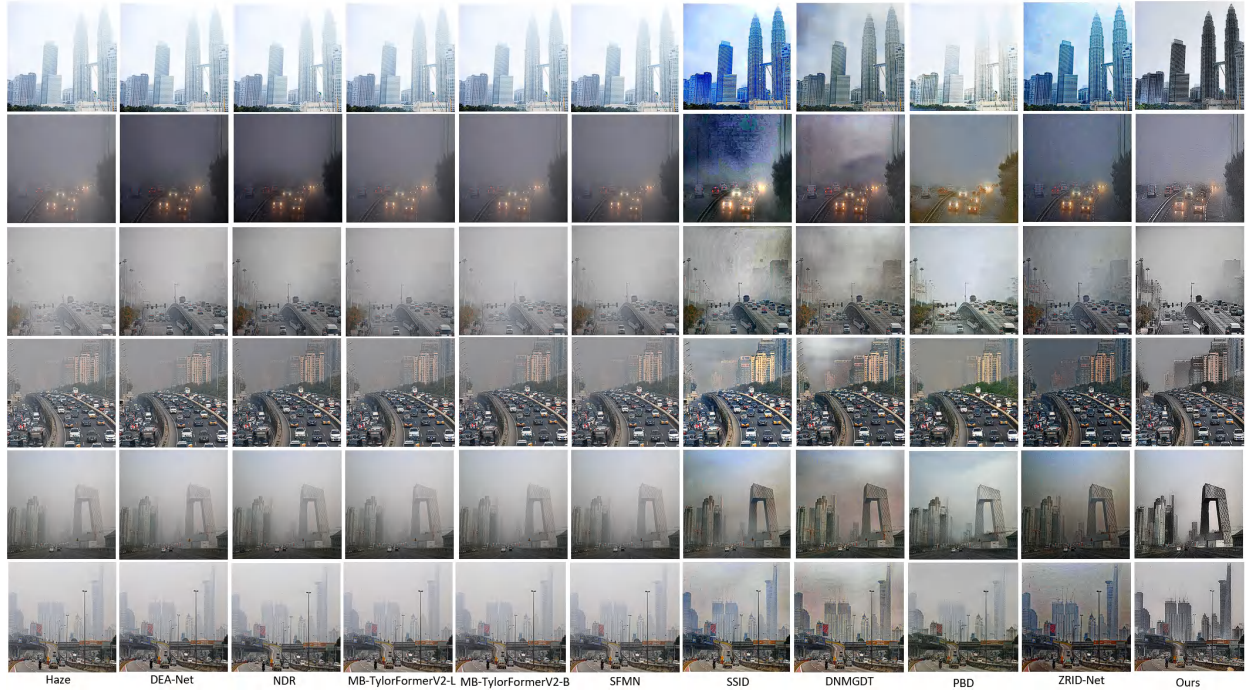


Figure 20. More qualitative comparison on the RTTS (Li et al., 2019) dataset. Zoom in for better visualization.

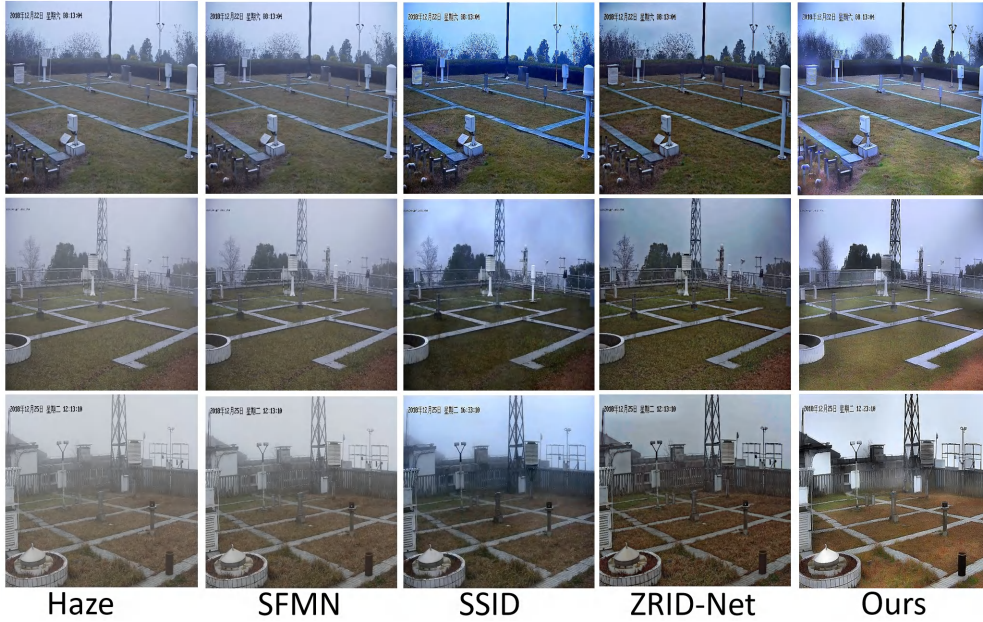


Figure 21. More qualitative comparison on the RW-HAZE (Chen et al., 2022) dataset.



Figure 22. Qualitative comparisons on nighttime hazy images.



Figure 23. Qualitative comparisons on dense haze images.

Hallucination: Missing intricate details on the car



Hallucination: Introducing artificial trees



Haze

w/o R

Ours

Figure 24. Visual results of the ablation study of R . The ablation of R may lead hallucinations.